

Mamba-CD: Mamba-Based Change Detection Network for Remote Sensing Images With Change Region-Aware Attention and Recursive Context Refinement Mechanism

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Abstract—Currently, remote sensing image change detection based on convolutional neural network and transformer has achieved significant advances. However, the existing methods often suffer from insufficient global dependency modeling capabilities or low computational efficiency. Furthermore, they struggle to handle complex background interference effectively, leading to imprecise detection of change region boundaries. To address these limitations, in this article, a Mamba-based CD network for binary CD tasks, named Mamba-CD, has been proposed, in which VMamba is adopted as the backbone, enabling efficient global dependency modeling with linear complexity. First, a change region-aware attention mechanism is designed to enhance features in change regions while suppressing features in background regions. The generated attention maps are constrained by a boundary loss function to strengthen the network's capability for the refined localization of change region boundaries. Subsequently, a gated recursive context refinement module is proposed. It utilizes a gating mechanism to filter out unimportant features and progressively fuses high-level global semantic information with low-level features in a recursive manner, thereby enhancing feature representation capability. Experimental results on three public datasets of LEVIR-CD, WHU-CD, and DSIFN-CD demonstrate that, compared to state-of-the-art CD methods, the proposed Mamba-CD achieves superior detection performance with lower model complexity and yields more precise boundaries of change regions.

Index Terms—Change region-aware attention, Mamba, recursive context refinement, remote sensing image change detection (RSICD), state space model (SSM).

I. INTRODUCTION

REMOTE sensing image change detection (RSICD) aims to identify changes in land cover by analyzing the same geographical area in remote sensing images acquired at different times. It is of great significance in many fields, such

as urban planning [1], agricultural resource management [2], disaster emergency response [3], and ecological environment protection [4]. However, remote sensing images often exhibit complex scenes, significant illumination and seasonal variations, background interference, and diverse shapes and scattered orientations of changed objects, posing substantial challenges for RSICD tasks.

In recent years, deep learning-based RSICD has achieved remarkable breakthroughs, significantly outperforming traditional methods. Deep models such as convolutional neural network (CNN) and transformer have been increasingly applied to RSICD tasks, leading to continuous improvement in detection performance. Nevertheless, these methods suffer from inherent limitations. First, CNN-based methods [5], [6], [7] employ fixed-size convolutional kernels to extract local features, resulting in limited receptive field. This makes it difficult for them to capture long-range dependencies, leading to information fragmentation when processing large-scale scenes. Incomplete or missed detection often occurs, particularly for scattered small target regions in complex scenarios.

Transformer excels at modeling long-range dependencies. Consequently, it has been applied to RSICD tasks [8], [9], [10], leveraging the self-attention mechanism to model long-range dependencies, which effectively captures global contextual information within images. However, the self-attention mechanism of transformer has quadratic computational complexity, which requires substantial computational resources when processing high-resolution remote sensing images. Furthermore, the global modeling tendency of transformer can diminish its ability to finely capture local details and boundaries in images.

Recently, Mamba [11], a state space model (SSM)-based approach for long-sequence modeling, has been proposed. Research results demonstrate that Mamba possesses the same excellent capability as transformer for capturing global contextual information, but with linear computational complexity.

Owing to these advantages, Mamba has been applied to RSICD tasks [12], [13], [14]. Chen et al. [12] pioneered the application of Mamba to RSICD, proposing ChangeMamba. The experimental results indicate that it achieves an effective tradeoff between detection performance and computational complexity. RSM-CD [15] improved Mamba's scanning strategy by

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introducing an omnidirectional selective scan module (OSSM) that selectively scans images in multiple directions, enhancing global modeling efficiency and consequently improving detection performance. However, these methods primarily focus on the efficient capture of global contextual information, while paying insufficient attention to the boundaries of change regions. This leads to difficulties in accurately detecting small change regions and obtaining fine boundaries in complex scenes.

To address the aforementioned issues, in this article, we propose a Mamba-based RSICD network, named Mamba-CD, which employs VMamba [13] as its backbone to efficiently model long-range dependencies with linear computational complexity.

Our main contributions are as follows.

- 1) VMamba is adopted as the backbone network to model long-range dependencies in remote sensing images, effectively capturing global contextual information with lower model computational complexity.
- 2) Change region-aware attention mechanism (CRAM) is proposed to enhance the features of change regions while suppressing those of background regions. Combined with boundary loss, it significantly strengthens the network's capability to refine the localization of change region boundaries.
- 3) Gated recursive context refinement module (GRCRM) is proposed, which adaptively determines fusion weights via a gating mechanism and dynamically integrates high-level semantic information with low-level detail information in a recursive manner, enhancing feature representation capability.

Experimental results on three public datasets of LEVIR-CD, WHU-CD, and DSIFN-CD demonstrate that, compared to the state-of-the-art CD methods, the proposed Mamba-CD achieves superior detection performance with lower model complexity and yields more precise boundaries of change regions.

The rest of this article is organized as follows. Section II introduces related research work. Section III details the proposed Mamba-CD network. Section IV reports the experimental results and analysis. Finally, Section V concludes this article.

II. RELATED WORKS

On the basis of the backbone networks adopted, the existing RSICD methods can be divided into three categories: CNN-based methods, transformer-based methods, and Mamba-based methods. Next, we will review the research advances on these methods.

CNN-based RSICD methods: In 2014, Daudt et al. [5] first introduced the fully convolutional network to the RSICD tasks. They proposed three fundamental architectures: Fully convolutional early fusion (FC-EF), which concatenates bi-temporal images along the channel dimension as input, and two Siamese-based variants, fully convolutional Siamese-concatenation (FC-Siam-Conc) and fully convolutional Siamese-difference. The experimental results demonstrate that the FC-EF architecture achieves superior detection performance, laying the foundation for subsequent research. Building upon this

framework, researchers adopted deeper and more powerful backbone networks, such as ResNet [16] and DenseNet [17], significantly enhancing the feature extraction capability of the backbone networks. Representative methods include ISNet [18] and SNUNet [7]. To improve feature representation capability, various attention mechanisms are also designed and utilized. DTCDSN [19] employs a dual-attention mechanism to enhance interdependencies between channels and space. STANet [20] proposes a self-attention mechanism based on spatiotemporal relationships, obtaining illumination-invariant features. DSAMNet [21] improves its deep-supervised network by introducing an attention mechanism to acquire more discriminative feature representation.

Despite significant progress, CNN-based methods' limited receptive field due to fixed-size convolutional kernels hinders their ability to capture long-range dependencies. This issue is particularly pronounced in scenes with complex contextual dependencies, often leading to missed or false detections of small target regions.

Transformer-based RSICD methods: The transformer, leveraging its powerful capability for modeling long-range dependencies, offers a solution to the challenge of capturing global contextual information faced by CNNs in RSICD tasks. In 2021, Chen et al. [10] pioneered the application of Transformer to RSICD, proposing the bitemporal image transformer (BIT) model. BIT utilizes ResNet to extract feature maps from bitemporal images, then treats these maps as "visual tokens" input to a transformer encoder for global self-attention modeling, effectively fusing the contextual information from both temporal images. ChangeFormer [9] employs a pure transformer-based encoder-decoder framework, performing self-attention aggregation in the spatiotemporal domain to directly contrast global contextual differences between the two temporal phases, enabling flexible capture of long-range dependencies between arbitrary pixel pairs. SwinSUNet [8] embeds weight-shared Swin Transformer [22] into the Siamese U-shaped framework to effectively extract multiscale global contextual information.

Although transformer-based methods exhibit clear advantages in modeling long-range dependencies, their computational complexity grows quadratically with the number of tokens, demanding substantial computational resources. Furthermore, transformers are often inadequate at capturing fine-grained local details. To balance global and local information, researchers have proposed hybrid CNN-Transformer architectures. MSTDSNet-CD [23] combines ResNet-18 with multiscale Swin Transformer to enhance the feature extraction. TransUNetCD [24] tokenizes feature maps extracted by the CNN and encoded them to extract rich global contextual information. However, these methods often achieve performance gains at the cost of increased computational complexity.

Mamba-based RSICD methods: Recent studies have demonstrated the effectiveness of SSMS [25], particularly the Mamba model [11], for long-sequence modeling. Mamba possesses the capability to model long-range dependencies while its computational complexity grows linearly or nearly linearly with the input sequence length. Mamba is now being widely applied to computer vision tasks, leading to various variants. Liu et al. [12]

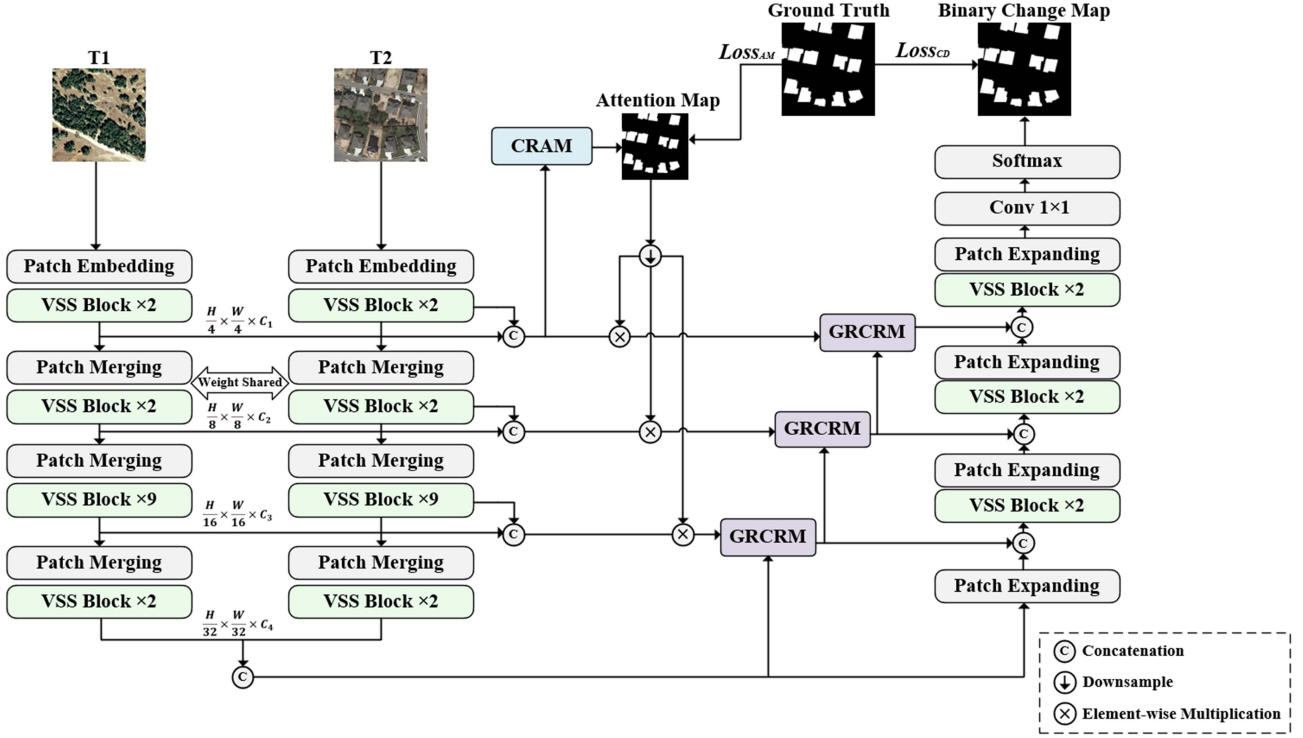


Fig. 1. Overall architecture of the proposed Mamba-CD.

proposed Vim, introducing positional encodings and a bidirectional scanning mechanism. Zhu et al. [13] proposed VMamba, which employs a cross-scanning strategy to process 2-D image data, successfully extending Mamba's application from 1-D sequences to 2-D images.

The potential of Mamba has also been rapidly explored in the field of remote sensing image processing. Wang et al. [26] proposed TrackingMamba, which pioneers the application of Mamba in single object tracking tasks. It achieves an efficient balance between accuracy and computational cost for uncrewed aerial vehicle tracking. Peng et al. [27] introduced Fusion-Mamba, featuring a dual-input Mamba block to effectively integrate spatial and spectral information. Similarly, Chen et al. [28] applied Mamba to infrared small-target detection with a Mamba-in-Mamba structure to capture both local and global features.

In 2024, ChangeMamba [14] pioneers the application of Mamba to RSICD tasks, using VMamba as an encoder to learn global spatial context from input images and introducing three spatiotemporal relationship modeling mechanisms in the decoder. RSM-CD [15] improves Mamba's scanning strategy by proposing an OSSM, extending VMamba's four-directional scanning to eight directions to capture spatial relationships more comprehensively. CWMamba [29] combines Mamba blocks with CNN to extract the features from the bitemporal images. CDMamba [30] attempts to integrate convolutional operations within the Mamba block to compensate for the limitations of pure Mamba architectures in extracting local details.

While these works effectively demonstrate Mamba's advantages in RSICD, they still exhibit limitations in capturing local details and refining the boundaries of change regions,

as well as in the efficient fusion of high-level and low-level features.

In this article, we adopt VMamba as the backbone network and propose a CRAM to suppress background noise interference and enhance the detection precision of change region boundaries. Furthermore, we design a GRCRM to adaptively fuse high-level semantic information with low-level detail information, ultimately achieving higher detection performance.

III. METHODOLOGY

A. Overall Architecture

The overall architecture of the proposed Mamba-CD is illustrated in Fig. 1. It primarily adopts a U-shape framework, consisting of a Mamba encoder-decoder. CRAM and GRCRM are embedded into it to boost the detection accuracy. The network takes bitemporal remote sensing images $T_1 \in \mathbb{R}^{H \times W \times 3}$ and $T_2 \in \mathbb{R}^{H \times W \times C}$ as input, where H and W denote the height and width of the images, respectively, and C represents the number of channels. The two images are first fed into two weight-shared Mamba encoders to extract multiscale features through four stages, denoted as E1 ~ E4. For the k th temporal image ($k \in \{1, 2\}$), the output features of each stage are denoted as $F_k^1 \in \mathbb{R}^{\frac{H}{4} \times \frac{W}{4} \times C_1}$, $F_k^2 \in \mathbb{R}^{\frac{H}{8} \times \frac{W}{8} \times C_2}$, $F_k^3 \in \mathbb{R}^{\frac{H}{16} \times \frac{W}{16} \times C_3}$, $F_k^4 \in \mathbb{R}^{\frac{H}{32} \times \frac{W}{32} \times C_4}$, respectively. Through hierarchical state space modeling, this encoder captures both local details and global contextual information at multiple scales.

Next, bitemporal features at the same scale are concatenated along the channel dimension to obtain features $\{F_i\}_{i=1}^4$. Considering that the shallow features of Mamba contain rich

low-level visual information, such as edges and contours, CRAM is applied to the shallow feature $\mathbf{F}_1$ to generate an attention map. This attention map is then reshaped to different scales via downsampling operations and multiplied with features $\{\mathbf{F}_i\}_{i=1}^3$ to enhance features of changed regions while suppressing those of background regions.

Subsequently, features from four stages are processed by the GRCRM. Guided by a gating mechanism, features are progressively fused from the deepest layer toward shallower layers, yielding enhanced multiscale features. Finally, the Mamba decoder receives these enhanced multiscale features via skip connections, gradually upsamples them to restore spatial details, and outputs a binary change map $\mathbf{Y} \in \mathbb{R}^{H \times W \times 1}$ after a linear projection layer and softmax normalization.

In the following sections, we will describe the implementation details of each part.

B. Mamba Encoder

Remote sensing images exhibit complex scenes with multi-scale and scattered objects, which demand a powerful backbone capable of efficiently modeling long-range dependencies. To this end, we employ Mamba [11], which is built upon the SSM [25], as our encoder. The SSM describes the evolution of a system's internal state over time. For a 1-D input sequence $x(t)$, the continuous state space equation is defined as

$$\begin{aligned} h'(t) &= \mathbf{A}h(t) + \mathbf{B}x(t) \\ y(t) &= \mathbf{C}h(t) \end{aligned} \quad (1)$$

where $h(t)$ represents the hidden state at time t , $\mathbf{A}$ is the state matrix governing the state transition, $\mathbf{B}$ is the input matrix, and $\mathbf{C}$ is the output matrix. To make the model compatible with discrete sequence data, the continuous equation is discretized using a timescale parameter Δ via the zero-order hold method, resulting in the discrete form

$$\begin{aligned} h_t &= \bar{\mathbf{A}}h_{t-1} + \bar{\mathbf{B}}x_t \\ y_t &= \mathbf{C}h_t \end{aligned} \quad (2)$$

where $\bar{\mathbf{A}} = e^{\Delta\mathbf{A}}$ and $\bar{\mathbf{B}} = (\Delta\mathbf{A})^{-1}(e^{\Delta\mathbf{A}} - \mathbf{I}) \cdot \Delta\mathbf{B}$. This allows the model to efficiently handle sequential data in a recurrent manner.

A key innovation of Mamba is its selective scanning mechanism. Unlike traditional SSMs with fixed parameters ($\mathbf{A}$, $\mathbf{B}$, Δ), Mamba makes these parameters input-dependent, enabling the model to selectively propagate or forget information based on the current context. This stands in contrast to the self-attention mechanism in transformers, which computes pairwise interactions between all tokens, resulting in $O(N^2)$ computational complexity for a sequence of length N . Mamba's selective mechanism maintains a linear complexity $O(N)$ with respect to sequence length while effectively capturing long-range dependencies.

In our architecture, the Mamba encoder comprises four stages to progressively extract multiscale image features. The first stage consists of patch embedding followed by two VSS blocks. Stages 2–4 each perform downsampling operation via patch merging,

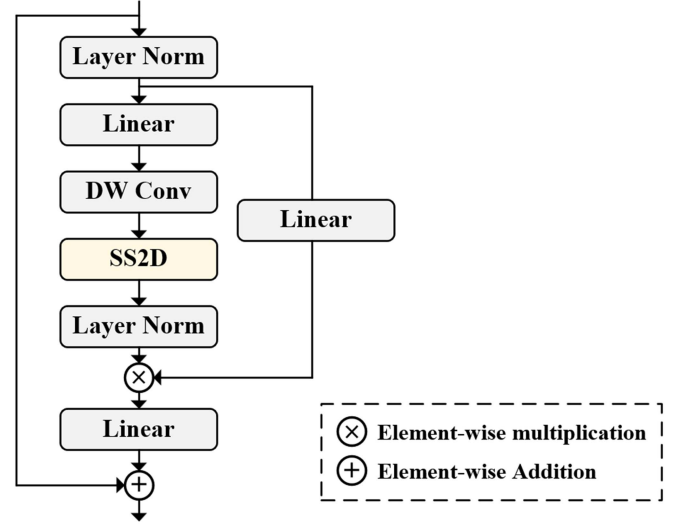


Fig. 2. Structure of the VSS block.

and then extract deeper semantic features through two VSS blocks.

Within each stage, the VSS block serves as the basic unit, its structure shown in Fig. 2. It enhances model stability and nonlinear representation capacity through residual connections. One branch performs feature extraction using a linear layer, activation layer, separable convolution, and a 2-D-selective-scan (SS2D) module, while the other branch utilizes a linear layer and an activation layer.

The SS2D module [13] is the core component of the VSS block. Its operation involves three steps: scan expanding, selective scanning spatial state sequence model (S6), and scan merging, as depicted in Fig. 3. The scan expanding unfolds the input image into four distinct directional sequences: top-left to bottom-right, bottom-right to top-left, top-right to bottom-left, and bottom-left to top-right. This scanning pattern ensures that any pixel integrates spatial contextual information from all four directions, achieving a global receptive field across multiple scan paths while maintaining linear computational complexity. These four sequences are then fed into the S6 block for 1-D state space modeling. Finally, the scan merging reconstructs the 1-D sequences back into a 2-D feature map for output. Given an input 2-D feature map $\mathbf{X} \in \mathbb{R}^{H \times W \times C}$, the SS2D process can be represented as

$$\mathbf{X}_v = \text{expand}(\mathbf{X}, v), \quad v \in V = \{1, 2, 3, 4\} \quad (3)$$

$$\hat{\mathbf{X}}_v = \mathbf{S}_6(\mathbf{X}_v) \quad (4)$$

$$\hat{\mathbf{X}} = \text{merge}(\hat{\mathbf{X}}_v \mid v \in V) \quad (5)$$

where v represents the four scanning directions, and $\text{expand}(\cdot)$ and $\text{merge}(\cdot)$ correspond to the scan expanding and scan merging operations, respectively.

Through hierarchical SSM, the Mamba encoder obtains features at four scales. Low-level features retain spatial details, such as edges, textures, and contours via local state modeling.

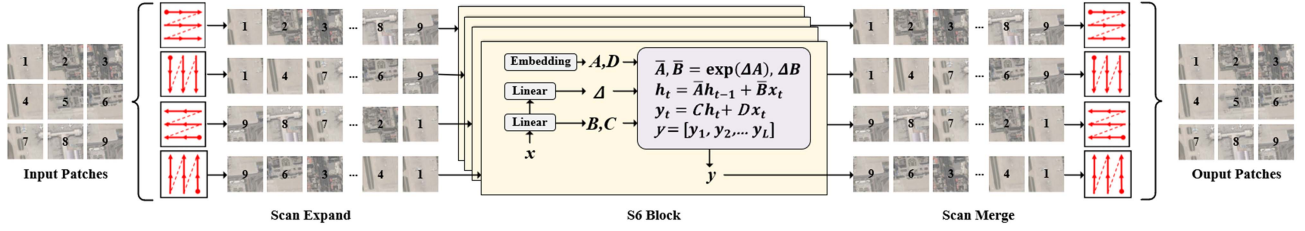


Fig. 3. Operation process of SS2D.

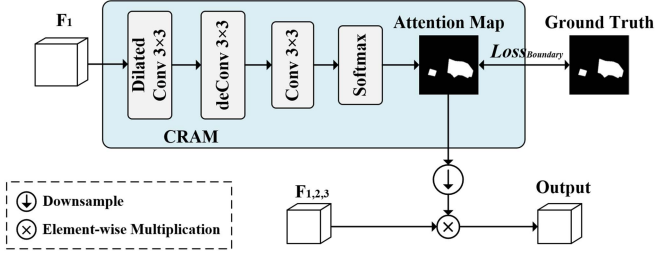


Fig. 4. Structure of the CRAM module.

High-level features contain long-range dependencies and contextual relationships. This enables Mamba to possess both local precision and global consistency in remote sensing images.

C. Change Region-Aware Attention Mechanism

In RSICD tasks, changed regions often suffer from blurred boundaries and background noise interference, especially under complex backgrounds, such as seasonal changes or shadow disturbances. This can lead the network to misclassify unchanged regions, having a negative impact on the detection accuracy. To address this issue, we propose CRAM to guide the network to focus more on the salient change regions and suppress the background noise interference. Simultaneously, boundary loss [31] is used to supervise the generation of the attention map, guiding the model to learn the boundary distribution of change regions for finer boundary detection.

CRAM is embedded after the first stage of the encoder. It utilizes information such as edges contained in the low-level features to generate an attention map, which is used to guide deeper features to enhance their response to change regions. The structure of CRAM is shown in Fig. 4, and its specific workflow is as follows.

For an input image $\mathbf{X} \in \mathbb{R}^{256 \times 256 \times 3}$, the fused feature output from the first encoder layer is $\mathbf{F}_1 \in \mathbb{R}^{64 \times 64 \times 3}$, which is fed into CRAM.

First, a 3×3 convolution with a dilation rate of 2 is conducted to extract spatial response features, followed by batch normalization (BN) and ReLU activation, yielding spatial feature $\mathbf{M}_1 \in \mathbb{R}^{64 \times 64 \times 64}$. This operation expands the receptive field, allowing the network to capture broader contextual information.

Then, a deconvolution operation is conducted to compress the channel dimension to produce $\mathbf{M}_2 \in \mathbb{R}^{64 \times 64 \times 2}$, a preliminary binary classification map for change and background. This is followed by 3×3 convolution, obtaining $\mathbf{M}_3 \in \mathbb{R}^{64 \times 64 \times 2}$.

Finally, softmax is applied along the channel dimension, and the change channel is extracted to obtain the attention map $\mathbf{A}_1 \in \mathbb{R}^{64 \times 64 \times 1}$, which represents the probability of each pixel belonging to the change class at the current scale.

To adapt to fusion features at different scales, this attention map is downsampled twice, obtaining attention maps $\mathbf{A}_2$ and $\mathbf{A}_3$. $\mathbf{A}_1$, $\mathbf{A}_2$, and $\mathbf{A}_3$ are then elementwise multiplied with the encoder outputs of $\mathbf{F}_1$, $\mathbf{F}_2$, and $\mathbf{F}_3$, respectively, yielding enhanced features $\tilde{\mathbf{F}}_1$, $\tilde{\mathbf{F}}_2$, and $\tilde{\mathbf{F}}_3$. This operation enhances change target regions while suppressing background regions across different scales. The entire process can be expressed by the following formulas:

$$\mathbf{M}_1 = \text{ReLU}(\text{BN}(\text{Conv}_{3 \times 3}(\mathbf{F}_1))) \quad (6)$$

$$\mathbf{M}_2 = \text{Deconv}_{3 \times 3}(\mathbf{M}_1) \quad (7)$$

$$\mathbf{M}_3 = \text{ReLU}(\text{BN}(\text{Conv}_{3 \times 3}(\mathbf{M}_2))) \quad (8)$$

$$\mathbf{A}_1 = \text{Softmax}(\mathbf{M}_3) \quad (9)$$

$$\mathbf{A}_2 = \text{AvgPool}_{2 \times 2}(\mathbf{A}_1) \quad (10)$$

$$\mathbf{A}_3 = \text{AvgPool}_{2 \times 2}(\mathbf{A}_2) \quad (11)$$

$$\tilde{\mathbf{F}}_i = \mathbf{F}_i \odot \mathbf{A}_i, \quad i \in \{1, 2, 3\} \quad (12)$$

where $\odot$ denotes elementwise multiplication.

Due to blurred boundaries and the small pixelwise proportion of change regions, traditional cross-entropy loss struggles to optimize the boundaries of change regions precisely. Therefore, we employ boundary loss to supervise the generation of attention map $\mathbf{A}$, constraining the spatial consistency between $\mathbf{A}$ and the ground truth (GT) through the boundary F1 score.

The calculation of boundary loss is as follows.

For the attention map $\mathbf{A}$ and GT $\mathbf{Y}^{\text{GT}} \in \{0, 1\}^{H \times W}$, boundary maps $\mathbf{B}_{\text{pred}}$ and $\mathbf{B}_{\text{GT}}$ are extracted via pooling operations. The boundaries are then dilated, yielding dilated boundaries $\mathbf{B}_{\text{pred}}^{\text{ext}}$ and $\mathbf{B}_{\text{GT}}^{\text{ext}}$. Based on the matching degree between the boundary maps, boundary precision P_b and boundary recall R_b are calculated as

$$P_b = \frac{\sum (\mathbf{B}_{\text{pred}} \odot \mathbf{B}_{\text{gt}}^{\text{ext}})}{\sum \mathbf{B}_{\text{pred}}} \quad (13)$$

$$R_b = \frac{\sum (\mathbf{B}_{\text{pred}}^{\text{ext}} \odot \mathbf{B}_{\text{gt}})}{\sum \mathbf{B}_{\text{gt}}} \quad (14)$$

The boundary loss is then defined as

$$\mathcal{L}_{\text{Boundary}} = 1 - 2 \cdot \frac{P_b R_b}{P_b + R_b} \quad (15)$$

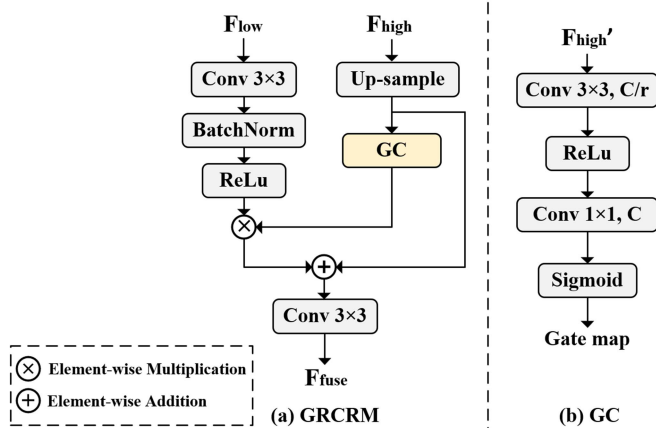


Fig. 5. Structure of the GRCRM and GC.

Boundary loss enables the network to focus on boundary pixels of change targets; therefore, the attention map may highlight the change regions while exhibiting finer boundary responses.

D. Gated Recursive Context Refinement Module

The features of deeper layer generally contain rich high-level semantic information, while those of the shallow layer contain low-level visual detailed information, such as edges and textures. Fusing features from different levels can enhance feature representation capability, thereby, improving detection performance. In this article, we propose the GRCRM. It recursively propagates high-level contextual information to the lower level features, achieving effective multiscale feature fusion.

In GRCRM, feature fusion is implemented through a three-stage recursive process. Given the three CRAM-enhanced features $\{\tilde{F}_i\}_{i=1}^3$, the recursive fusion proceeds as follows: the high-level feature F_4 and its adjacent lower level feature $\tilde{F}_3$ are input into the GRCRM to obtain the fused feature G_3 . Then, G_3 and $\tilde{F}_2$ are input into GRCRM to obtain G_2 . This process repeats, finally yielding the output G_1 , which incorporates the lowest level feature. The fused features $\{F_4, G_3, G_2, G_1\}$ serve as inputs to each decoder layer. This process is formulated as

$$G_3 = \text{GRCRM}(F_4, \tilde{F}_3) \quad (16)$$

$$G_{i-1} = \text{GRCRM}(G_i, \tilde{F}_{i-1}), \quad i \in \{2, 3\}. \quad (17)$$

The structure of GRCRM is shown in Fig. 5(a). It takes a high-level feature $\tilde{F}_i$ and a low-level feature $\tilde{F}_{i-1}$ as inputs. The specific workflow is as follows.

First, $\tilde{F}_i$ is bilinearly upsampled to $\tilde{F}_i'$, aligning spatial resolution with the low-level feature. $\tilde{F}_i'$ is then fed into the gating cell (GC) to generate a spatial gating map G . The low-level feature is enhanced via a 3×3 convolution, BN, and ReLU, resulting in $\hat{F}_{i-1}$. This enhanced feature is then elementwise multiplied with G , obtaining the weighted low-level feature W_{i-1} .

Finally, W_{i-1} is added to the upsampled high-level feature $\tilde{F}_i'$, and fused through a 3×3 convolution, yielding the fused feature G_{i-1} . The entire process can be expressed as

$$\tilde{F}_i' = \text{Upsample}(\tilde{F}_i) \quad (18)$$

$$G = \text{GC}(\tilde{F}_i') \quad (19)$$

$$\hat{F}_{i-1} = \text{ReLU}(\text{BN}(\text{Conv}_{3 \times 3}(\tilde{F}_{i-1}))) \quad (20)$$

$$W_{i-1} = \hat{F}_{i-1} \odot G \quad (21)$$

$$G_{i-1} = \text{Conv}_{3 \times 3}(W_{i-1} + \tilde{F}_i'). \quad (22)$$

To achieve effective fusion between features of different levels, a core module named the GC is designed within GRCRM, its structure shown in Fig. 5(b). After inputting the high-level features into GC, the channel number is first compressed to 1/4 via a 3×3 convolution. Following ReLU activation, a 1×1 convolution is conducted to restore the original channel number. Finally, a sigmoid function is exploited to generate a gating weight map with values between 0 and 1. This process is formulated as

$$G = \sigma(\text{Conv}_{1 \times 1}(\text{ReLU}(\text{Conv}_{3 \times 3}(\tilde{F}_i')))). \quad (23)$$

Based on the content of the high-level semantic features, GC adaptively generates a fusion weight map. This map dynamically adjusts the weighting coefficients between high-level and low-level features at different spatial positions, enabling dynamic fusion across levels.

E. Mamba Decoder

The Mamba decoder progressively upsamples the fused multiscale features back to the original resolution. Finally, a binary change map is generated via softmax normalization and thresholding. The decoder consists of four upsampling stages that gradually restore spatial information from deep, semantic features to shallow, detailed features.

In stages 1–3, the input feature maps are first upsampled by a factor of two via patch expanding, which enlarges the spatial resolution while halving the number of channels. The upsampled features are then fused with the skip-connected features from the GRCRM at the same scale. The feature fusion employs a channel concatenation strategy, which concatenates the upsampled features with the skip connection features along the channel dimension. This is followed by a linear projection layer that compresses the concatenated channels back to the original dimensionality, ensuring dimension compatibility while promoting effective information integration. The fused features are subsequently fed into two consecutive VSS blocks for further processing. The structure of these VSS blocks is identical to that used in the Mamba encoder, so as to capture of long-range dependencies through state space modeling.

In stage 4, the decoder performs the final upsampling and prediction operations. The features are upsampled by a factor of four to the original resolution via patch expanding. A 1×1 convolution then projects the feature maps into a two-channel output, corresponding to the “change” and “nonchange” classes, respectively. Finally, softmax is applied along the channel dimension, obtaining the probability distribution for each pixel, and a binary change map is finally generated through thresholding.

F. Loss Function

In RSICD tasks, change regions often occupy a small area, leading to class imbalance, and their boundaries are blurry and susceptible to background interference. So we jointly employ three loss functions to supervise the network training: Dice loss [32] and binary cross-entropy (BCE) loss [33] act directly on the final change detection output, measuring the results at the region level and pixel level, respectively. Boundary loss serves as a deep supervision signal for CRAM, constraining the boundary pixels of change regions on the attention map generated by this module.

Dice loss optimizes the segmentation quality of the overall region under class imbalance by measuring the overlap between prediction and GT. It is calculated as

$$\mathcal{L}_{\text{Dice}}(y, \hat{y}) = 1 - \frac{2 \cdot y \cdot \hat{y}}{y + \hat{y}} \quad (24)$$

where y and $\hat{y}$ represent the GT and the corresponding predicted change map, respectively.

BCE loss provides pointwise penalties on the class probability of each pixel, ensuring good classification capability at the pixel level. It is calculated as

$$\mathcal{L}_{\text{BCE}}(y, \hat{y}) = -(y \cdot \log(\hat{y}) + (1 - y) \cdot \log(1 - \hat{y})) \quad (25)$$

Boundary loss specifically constrains pixels at the edges of change regions. It supervises the attention map generated by CRAM to learn precise boundary distributions, thereby, improving the detection accuracy of change target boundaries. Its calculation formula is given by (13).

The total loss is formulated as

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{Dice}} + \mathcal{L}_{\text{BCE}} + \lambda \cdot \mathcal{L}_{\text{Boundary}} \quad (26)$$

where λ is a parameter balancing the influence of boundary supervision on the overall loss. Its value will be determined through subsequent experimental analysis.

IV. EXPERIMENT RESULTS AND ANALYSIS

To validate the effectiveness of the proposed Mamba-CD, we conducted experiments on three public CD datasets. The experimental results and analysis are presented as follows.

A. Datasets and Evaluation Metrics

Three public datasets were employed in the experiments: LEVIR-CD, WHU-CD, and DSIFN-CD.

LEVIR-CD [20]: It is a large-scale remote sensing building CD dataset, containing 637 pairs of Google Earth images. Each image has a size of 1024×1024 pixels with a spatial resolution of 0.5 m/pixel. The time span between the bitemporal images ranges from 5 to 14 years. Following the mainstream methods ([9], [10], [15], [30], [34]), each image was cropped into nonoverlapping patches of size 256×256 , resulting in 7120/1024/2048 image pairs for training/validation/testing, respectively.

WHU-CD [35]: It is a remote sensing CD dataset released by Wuhan University. The dataset consists of two aerial images with a size of $32\,507 \times 15\,354$ pixels and a spatial

resolution of 0.2 m/pixel. Following the mainstream methods ([10], [19], [30], [34], [36]), each image was cropped into nonoverlapping patches of size 256×256 , resulting in 6096/762/762 image pairs for training/validation/testing, respectively.

DSIFN-CD [6]: It is a public CD dataset manually collected from Google Earth, comprising six high-resolution images from different cities in China. Following the mainstream methods ([9], [37], [38]), each image was cropped into nonoverlapping patches of size 256×256 , resulting in 14 400/1360/192 image pairs for training/validation/testing, respectively.

Evaluation metrics: To comprehensively evaluate the performance of the proposed Mamba-CD, we adopted five commonly used CD performance metrics: Precision (Pre), recall (Rec), F1 score (F1), intersection over union (IoU), and overall accuracy (OA). The F1 score is a comprehensive metric combining precision and recall, providing a more holistic assessment of detection performance. IoU represents the overlap ratio between the detection results and the GT. OA indicates the proportion of correctly predicted pixels among all pixels. These metrics are defined as follows:

$$\text{Pre} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (27)$$

$$\text{Rec} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (28)$$

$$\text{F1} = \frac{2 \times \text{Pre} \times \text{Rec}}{\text{Pre} + \text{Rec}} \quad (29)$$

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \quad (30)$$

$$\text{OA} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \quad (31)$$

where TP, TN, FP, and FN denote the number of true positives, true negatives, false positives, and false negatives, respectively. Their values range from 0 to 1, with higher values indicating better performance.

B. Parameter Settings

The network model was built, trained, and tested on the PyTorch platform. The experiments were run on a computer with an Intel Core i7-11700 K CPU (3.60 GHz) and a Quadro RTX 6000 GPU. The AdamW optimizer was used with an initial learning rate of $1e-4$ and weight decay of $1e-3$. A dynamic learning rate decay strategy was employed: if the F1 score on the validation set did not improve for 10 consecutive epochs, the learning rate was reduced to 0.1 times its current value. The batch size was set to 16, and the total number of epochs was 200. Automatic mixed precision training was used to accelerate computation. The Mamba encoder was initialized with weights pretrained on the ImageNet dataset.

Input images were uniformly cropped into nonoverlapping patches of size 256×256 pixels. For the LEVIR-CD dataset, 7120 pairs were used for training, 1024 pairs for validation, and 2048 pairs for testing. For the WHU-CD dataset, 6096 pairs were used for training, 762 pairs for validation, and 762 pairs

TABLE I
QUANTITATIVE COMPARISON ON THE LEVIR-CD DATASET

Method	Year	Pre (%)	Rec (%)	F1 (%)	IoU (%)	OA (%)
FC-EF [5]	2018	86.91	80.17	83.40	71.53	98.39
FC-Siam-Di [5]	2018	89.53	83.31	86.31	75.92	98.67
FC-Siam-Conc [5]	2018	91.99	76.77	83.69	71.96	98.49
DTCDCN [19]	2020	88.53	86.83	87.67	78.05	98.77
STANet [20]	2020	83.81	91.00	87.26	77.40	98.66
IFNet [6]	2020	94.02	82.93	88.13	78.77	98.87
SNUNet [7]	2021	89.18	87.17	88.16	78.83	98.82
BIT [10]	2021	89.24	89.37	89.31	80.68	98.92
ChangeFormer [9]	2022	92.05	88.80	90.40	82.48	99.04
BiFA [34]	2024	91.52	89.86	90.69	82.96	99.06
RSM-CD [15]	2024	92.52	89.73	91.10	83.66	/
CDMamba [30]	2025	91.43	90.08	90.75	83.07	99.06
Ours	2025	93.06	91.07	92.06	85.28	99.20

TABLE II
QUANTITATIVE COMPARISON ON THE WHU-CD DATASET

Method	Year	Pre (%)	Rec (%)	F1 (%)	IoU (%)	OA (%)
FC-EF [5]	2018	71.63	67.25	69.37	53.11	97.61
FC-Siam-Di [5]	2018	47.33	77.66	58.81	41.66	95.63
FC-Siam-Conc [5]	2018	60.88	73.58	66.63	49.95	97.04
DTCDCN [19]	2020	63.92	82.30	71.95	56.19	97.42
STANet [20]	2020	79.37	85.50	82.32	69.95	98.52
IFNet [6]	2020	96.91	73.19	83.40	71.52	98.83
SNUNet [7]	2021	85.60	81.49	83.50	71.67	98.71
ChangeFormer [9]	2022	91.83	88.02	89.88	81.63	99.12
BiFA [34]	2024	95.15	93.60	94.37	89.34	99.56
RSM-CD [15]	2024	93.37	90.42	91.87	84.96	/
SChanger [36]	2025	94.62	91.83	93.20	87.27	/
CDMamba [30]	2025	95.58	92.01	93.76	88.26	99.51
Ours	2025	96.52	93.91	95.20	90.83	99.62

for testing. For the DSIFN-CD dataset, 14 400 pairs were used for training, 1360 pairs for validation, and 192 pairs for testing.

C. Comparison With SOTA Methods

To validate the effectiveness of the proposed Mamba-CD, we compared it with 15 existing SOTA methods, including CNN-based methods (FC-EF [5], FC-Siam-Di [5], FC-Siam-Conc [5], DTCDCN [19], STANet [20], IFNet [6], SNUNet [7], SChanger [36], ADSFNet [38]), transformer-based methods (ChangeFormer [9]), CNN+Transformer-based methods (BIT [10], FTAN [37], BiFA [34]), and Mamba-based methods (RSM-CD [15], CDMamba [30]). Data for those methods other than ours were cited from the literatures.

1) *Quantitative Comparison*: Tables I–III present the quantitative comparison between our proposed Mamba-CD and other methods on the LEVIR-CD, WHU-CD, and DSIFN-CD datasets, respectively. The best and second-best results are highlighted in red and blue, respectively. It can be observed that our proposed network achieves the best or second-best performance on all three datasets. Detailed analysis of the experimental results on each dataset is provided as follows.

On the LEVIRCD dataset, Mamba-CD achieved the best values in terms of recall, F1, IoU, and OA metrics, and the second-best result in precision. Compared to the second-best method, F1 and IoU values were improved by 0.96% and 1.62%, respectively, recall and OA were 0.07% and 0.14% higher than the second-best values, while also demonstrating strong competitiveness in precision.

On the WHUCD dataset, Mamba-CD also achieved the best results in terms of recall, F1, IoU, and OA metrics, and the second-best result in precision. Compared to the second-best method, precision, recall, F1, IoU, and OA values were improved by 1.37%, 0.31%, 0.83%, 1.49%, and 0.06%, respectively.

On the DSIFNCD dataset, Mamba-CD achieved the best results in all five metrics. Compared to the second-best method, improvements of 0.81%, 0.37%, 0.60%, 1.19%, and 0.21% were achieved, respectively.

Notably, when compared to RSM-CD and CDMamba, which are also built on the Mamba architecture, our proposed Mamba-CD demonstrates consistent improvements across all evaluation metrics. On the LEVIR-CD dataset, Mamba-CD achieves 0.96% and 1.31% higher F1 and 1.62% and 2.21% higher IoU metric than RSM-CD and CDMamba, respectively. A similar trend can

TABLE III
QUANTITATIVE COMPARISON ON THE DSIFN-CD DATASET

Method	Year	Pre (%)	Rec (%)	F1 (%)	IoU (%)	OA (%)
FC-EF [5]	2018	72.61	52.73	61.09	43.98	88.59
FC-Siam-Di [5]	2018	59.67	65.71	62.54	45.50	86.63
FC-Siam-Conc [5]	2018	66.45	54.21	59.71	42.56	87.57
DTCDCSCN [19]	2020	53.87	77.99	63.72	46.76	84.91
STANet [20]	2020	67.71	61.68	64.56	47.66	88.49
IFNet [6]	2020	67.86	53.94	60.10	42.96	87.83
SNUNet [7]	2021	60.60	72.89	66.18	49.45	87.34
ChangeFormer [9]	2022	88.48	84.94	86.67	76.48	95.56
BiFA [34]	2024	73.99	68.87	71.34	55.45	90.80
FTAN [37]	2024	90.54	88.61	89.56	81.10	-
ADSFNet [38]	2025	94.79	95.24	95.01	90.50	98.30
Ours	2025	95.60	95.61	95.61	91.69	98.51

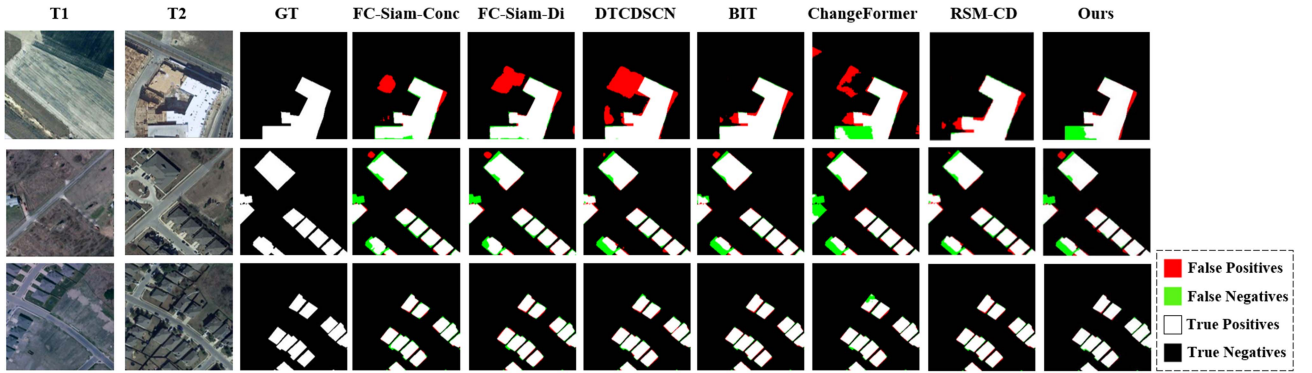


Fig. 6. Visualization results of different methods on the LEVIR-CD dataset.

be observed on the WHU-CD dataset that, Mamba-CD shows superior performance over both RSM-CD and CDMamba with margins of 3.33% and 1.44% in F1, and 5.87% and 2.57% in IoU metric, respectively. These performance gains, particularly in F1 and IoU metric, are owing to the CRAM, which enhances change region features and suppresses background interference, complemented by the GRCRM, which enables recursive multiscale feature fusion for more complete and precise change detection.

Synthesizing the above-mentioned quantitative comparison results, Mamba-CD consistently achieved the best or second-best detection performance across all three datasets, fully validating its effectiveness.

2) *Qualitative Comparison*: To visually demonstrate the detection results, we visualized the detection results produced by different methods. Fig. 6 shows the visualization results of the samples of the LEVIR-CD dataset. The first and second columns display the bitemporal images, and the third column shows the GT. The fourth to the last columns show the visualization results of FC-Siam-Conc, FC-Siam-Di, DTCDCSCN, BIT, ChangeFormer, RSM-CD, and our method, respectively. Different colors are used to annotate pixels in the visualization results: red indicates FP, green indicates FN, white indicates TP, and black indicates TN.

It can be observed that the detection results from FCSiam-Conc, FCSiamDi, DTCDCSCN, and ChangeFormer exhibit large

areas of false detections in changed regions (marked in red), whereas Mamba-CD shows almost none. Furthermore, when compared to RSM-CD, which is also based on the Mamba architecture, our Mamba-CD yields change regions with fine boundaries and significantly fewer FPs in the background.

This is primarily because Mamba-CD can perform global state space modeling, ensuring the network's perception of large-scale change regions. Simultaneously, the proposed CRAM specifically enhances features in potential change regions while suppressing responses in background areas. Consequently, our proposed Mamba-CD effectively suppresses “pseudochanges” and reduces the missed detection rate. Mamba-CD also shows significantly fewer false detections and missed detections at the boundaries of change regions compared to other methods. This indicates that our proposed Mamba-CD can not only effectively capture global features but also precisely delineate the boundaries of change regions.

3) *Comparison of Computational Complexity and Parameter Scale*: To comprehensively evaluate model efficiency and performance, we compared Mamba-CD with various advanced methods on the LEVIR-CD dataset in terms of computational complexity, parameter scale, and detection metrics, as shown in Table IV. Mamba-CD has 27.94 million (M) parameters and 13.08 Giga (G) floating point operations (FLOPs). Compared to RSMCD, which also uses the Mamba architecture, Mamba-CD

TABLE IV
COMPARISON OF COMPUTATIONAL COMPLEXITY AND PARAMETER SCALE WITH OTHER METHODS

Method	Params. (M)	FLOPs (G)	Pre (%)	Rec (%)	F1 (%)	IoU (%)	OA (%)
FC-Siam-Di [5]	1.34	4.72	89.53	83.31	86.31	75.92	98.67
FC-Siam-Conc [5]	1.54	5.33	91.99	76.77	83.69	71.96	98.49
STANet [20]	16.93	6.58	83.81	91.00	87.26	77.40	98.66
IFNet [6]	50.71	41.10	94.02	82.93	88.13	78.77	98.87
SNUNet [7]	12.03	27.44	89.18	87.17	88.16	78.83	98.82
ChangeFormer [9]	41.03	101.57	92.05	88.80	90.40	82.48	99.04
RSM-CD [15]	27.90	15.70	92.52	89.73	91.10	83.66	/
Ours	27.94	13.08	93.06	91.07	92.06	85.28	99.20

TABLE V
ABLATION STUDIES ON THE LEVIR-CD DATASET

Baseline	CRAM	GRCRM	Pre (%)	Rec (%)	F1 (%)	IoU (%)	OA (%)
✓			92.48	90.58	91.52	84.37	99.14
✓	✓		92.59	91.02	91.81	84.85	99.17
✓		✓	92.99	90.88	91.92	85.07	99.19
✓	✓	✓	93.06	91.07	92.06	85.28	99.20

maintains a comparable parameter scale while reducing computational cost by 16.7%. Compared to the transformer-based ChangeFormer, Mamba-CD uses only 68% of its parameters and 13% of its FLOPs. Compared to CNN-based architectures, such as SNUNet and STANet, although Mamba-CD has a slightly larger parameter scale, its detection accuracy far surpasses these two methods. Compared to RSM-CD, which has a comparable parameter scale, Mamba-CD improves precision, recall, F1, and IoU values by 0.54%, 1.34%, 0.96%, and 1.62%, respectively.

The above-mentioned comparison results demonstrate that Mamba-CD achieves superior detection performance while maintaining a moderate model size and efficient inference, striking a balance between computational efficiency and detection performance.

D. Ablation Studies

1) Effectiveness of the Proposed Components in Mamba-CD:

To validate the effectiveness of the proposed components in Mamba-CD, we conducted ablation experiments on the LEVIR-CD dataset. The results are presented in Table V, where the baseline (first row) represents the network only using the VMamba. The second and third rows show the results after incorporating the CRAM and GRCRM into the baseline, respectively. The final row indicates the performance when both components are added simultaneously, representing the complete Mamba-CD.

The results demonstrate that, incorporating the CRAM improves recall by 0.44%, indicating its ability to guide the network to focus more on the genuine change regions, suppress background noises, and refine boundary detection, thereby, reducing missed and false detections. Adding GRCRM boosts precision value by 0.51%, with F1 and IoU values increasing by 0.4% and 0.7%, respectively. This validates GRCRM's role in fusing high-level semantic information and low-level details to preserve genuine changes while reducing false alarms.

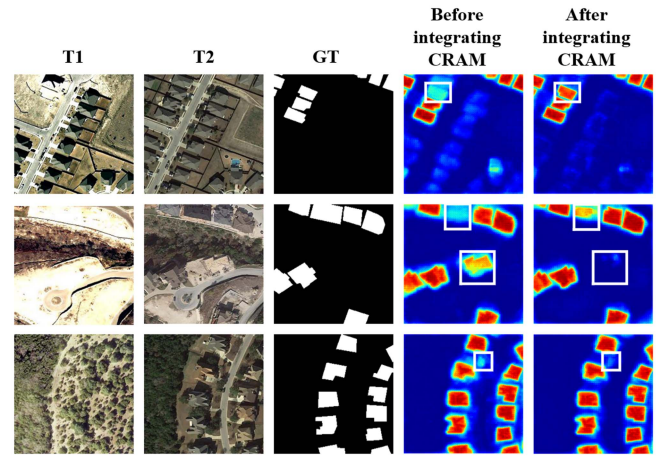


Fig. 7. Heatmaps before and after integrating CRAM on the LEVIR-CD dataset.

When both CRAM and GRCRM are added to the baseline, they exhibit synergistic effects. CRAM significantly enhances recall by amplifying responses to change regions, while GRCRM improves precision through multiscale feature fusion. Consequently, all detection metrics of Mamba-CD are elevated.

To further validate CRAM's effectiveness, we performed a comparative visual analysis of heatmaps and detection results before and after its integration, as shown in Figs. 7 and 8, respectively.

In Fig. 7, columns 1 and 2 display the bitemporal images, column 3 shows the GT, while columns 4 and 5 present the heatmaps before and after integrating CRAM. It is evident that without CRAM, the network exhibits insufficient attention to some genuine change regions (rows 1 and 2) and erroneous attention to pseudochange regions (rows 2 and 3). After integrating CRAM, responses in genuine change regions are significantly enhanced, while attention to pseudochange regions

TABLE VI
RESULTS OF WEIGHTING DIFFERENT ENCODER FEATURES LAYERS WITH CRAM

Stage 1	Stage 2	Stage 3	Stage 4	Pre (%)	Rec (%)	F1 (%)	IoU (%)	OA (%)
✓				92.88	91.03	91.94	85.09	99.19
✓	✓			92.92	91.06	91.99	85.15	99.19
✓	✓	✓		93.06	91.07	92.06	85.28	99.20
✓	✓	✓	✓	92.84	90.85	91.83	84.90	99.18

TABLE VII
RESULTS OF USING DIFFERENT FEATURE FUSION METHODS

Feature fusion method	Pre (%)	Rec (%)	F1 (%)	IoU (%)	OA (%)
Element-wise addition	92.49	90.99	91.73	84.73	99.16
Concatenation	92.78	90.85	91.81	84.85	99.17
Element-wise multiplication	92.95	90.80	91.87	84.95	99.18
Gating mechanism	93.06	91.07	92.06	85.28	99.20

TABLE VIII
RESULTS OF USING DIFFERENT LOSS FUNCTIONS

Loss function	Pre (%)	Rec (%)	F1 (%)	IoU (%)	OA (%)
Dice + BCE + SSIM + Boundary loss	91.89	90.84	91.51	84.10	99.13
Dice + BCE + IoU + Boundary loss	92.97	90.73	91.84	84.90	99.17
Dice + BCE + SSIM + IoU + Boundary loss	92.02	90.91	91.69	84.26	99.15
Dice + BCE + Boundary loss	93.06	91.07	92.06	85.28	99.20

TABLE IX
RESULTS OF USING DIFFERENT LOSS FUNCTION COEFFICIENTS

λ	Pre (%)	Rec (%)	F1 (%)	IoU (%)	OA (%)
0.3	93.06	91.07	92.06	85.28	99.20
0.5	93.04	90.95	91.98	85.16	99.19
0.7	92.92	90.82	91.86	84.94	99.18
1.0	93.21	90.71	91.95	85.09	99.19

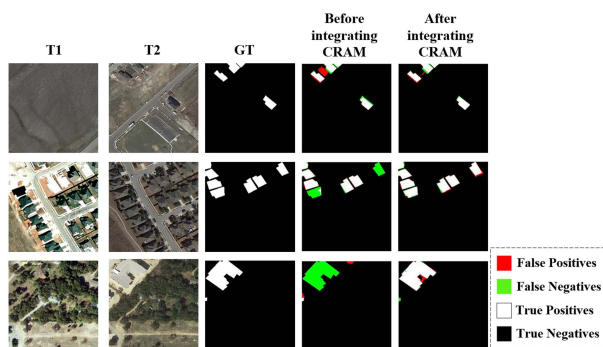


Fig. 8. Visualized detection results before and after integrating CRAM on the LEVIR-CD dataset.

is markedly suppressed. Furthermore, across all three example groups, attention distribution becomes more concentrated on genuine changes with sharper boundary contours after CRAM integration, compared to the more dispersed, wider light-blue coverage observed without CRAM.

In the visualization of detection results (see Fig. 8), the column arrangement follows Fig. 7, with columns 4 and 5 showing detection results before and after integrating CRAM. Without CRAM, the network exhibits significant FP in unchanged areas (row 1) and substantial missed detections in genuine change regions (rows 2 and 3). After integrating CRAM, FP are effectively eliminated, and missed regions are largely filled. Moreover, in all three comparison groups, the boundaries of change regions appear sharper after CRAM integration, with noticeably reduced FP and missed detections.

These visualization results further corroborate CRAM's efficacy in precisely focusing on change regions, suppressing background noise interference, and enhancing detection boundaries.

2) *Impact of CRAM Weighting on Different Feature Layers:* We evaluated CRAM's weighting effect across different encoder stages, and the results on LEVIR-CD are shown in Table VI. Optimal detection performance is achieved when features from stages 1–3 are weighted simultaneously. Performance degrades when CRAM is also applied to stage 4 features. This indicates that weighting mid- and low-level features effectively leverages

the attention map, while weighting highly abstract high-level features harms performance. Thus, we weight features from the first three stages only.

3) *Impact of Different Feature Fusion Strategies on the Detection Performance:* GRCRM uses GC for dynamic feature fusion. To assess the impact of different fusion strategies on the detection performance, we compared four fusion methods on the LEVIR-CD dataset: elementwise addition, channel concatenation, elementwise multiplication, and GC. As Table VII shows, GC outperforms other methods across metrics. This is attributed to GC's adaptive weight allocation based on semantic and detail information, enhancing feature representation.

4) *Impact of Different Loss Functions on the Detection Performance:* We conducted comparative experiments on the LEVIR-CD dataset using dice loss, BCE loss, Structural Similarity (SSIM) loss, IoU loss, and boundary loss. Results are shown in Table VIII. The combination of dice loss, BCE loss, and boundary loss achieves optimal performance across all five metrics. While adding IoU loss can boost precision and IoU to 92.97% and 84.90%, respectively, it reduces recall to 90.73%. Adding SSIM loss can increase recall value to 90.91%, but compromise precision value. Thus, we adopt dice loss, BCE loss, and boundary loss for joint supervision of network training.

5) *Impact of Loss Function Coefficient λ on the Detection Performance:* To investigate the influence of the boundary loss coefficient λ on the detection performance of Mamba-CD, experiments were conducted on the LEVIR-CD dataset, with results shown in Table IX. It can be observed that, when λ is set to 0.3, the model achieves the optimal balance between precision and recall, along with the highest F1 and IoU. Increasing λ beyond this point leads to a decline in both F1 score and IoU. Considering all metrics, λ is set to 0.3 in this article.

V. CONCLUSION

This article proposes Mamba-CD, an SSM Mamba-based network, to address challenges in existing RSICD methods: balancing global and local information, and precisely localizing change targets and boundaries. Through comprehensive experiments, the following conclusions can be drawn.

- 1) Mamba can achieve multiscale global context and local detail feature extraction with linear complexity.
- 2) The proposed CRAM enables the network to focus more effectively on change regions and boundaries, enhancing detection performance. Adding boundary loss as a constraint further enables the refinement of boundary detection.
- 3) For the GRCRM, employing GC to adaptively generate weighting coefficients yields better detection performance.

However, this work still has certain limitations. In some complex scenarios, the detection accuracy for small targets has room for further improvement. In addition, although the Mamba encoder-decoder possesses the advantage of linear complexity, the overall computational overhead remains substantial when processing ultrahigh-resolution remote sensing images. This poses challenges for deployment in very large-scale scenarios

or real-time applications. In future work, we will design a lightweight Mamba network to achieve desirable detection performance while reducing model parameters, aiming to meet the demands of resource-constrained environments and real-time applications.

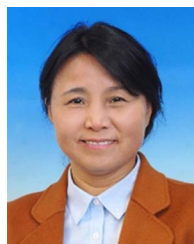
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